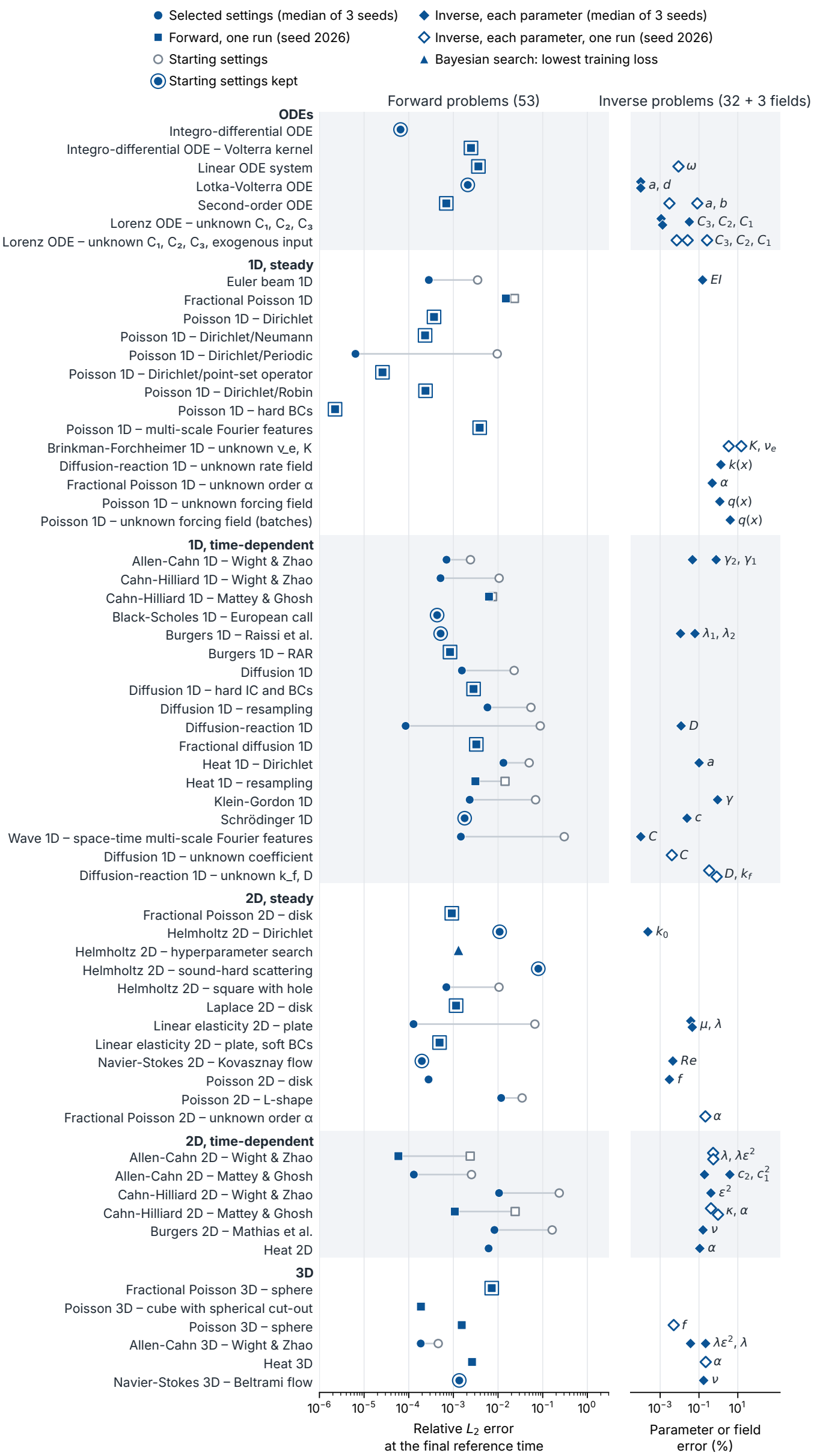

Selected settings (median of 3 seeds)
Forward, one run (seed 2026)
Starting settings
Starting settings kept
Inverse, each parameter (median of 3 seeds)
Inverse, each parameter, one run (seed 2026)
Bayesian search: lowest training loss
Forward problems (53)
Inverse problems (32 + 3 fields)
ODEs
Integro-differential ODE
Integro-differential ODE – Volterra kernel
Linear ODE system
Lotka-Volterra ODE
Second-order ODE
Lorenz ODE – unknown C_1, C_2, C_3
Lorenz ODE – unknown C_1, C_2, C_3, exogenous input
ω
a, d
a, b
C_3, C_2, C_1
C_3, C_2, C_1
1D, steady
Euler beam 1D
Fractional Poisson 1D
Poisson 1D – Dirichlet
Poisson 1D – Dirichlet/Neumann
Poisson 1D – Dirichlet/Periodic
Poisson 1D – Dirichlet/point-set operator
Poisson 1D – Dirichlet/Robin
Poisson 1D – hard BCs
Poisson 1D – multi-scale Fourier features
Brinkman-Forchheimer 1D – unknown ν_e, K
Diffusion-reaction 1D – unknown rate field
Fractional Poisson 1D – unknown order α
Poisson 1D – unknown forcing field
Poisson 1D – unknown forcing field (batches)
EI
K, ν_e
$k(x)$
α
$q(x)$
$q(x)$
1D, time-dependent
Allen-Cahn 1D – Wight & Zhao
Cahn-Hilliard 1D – Wight & Zhao
Cahn-Hilliard 1D – Mattey & Ghosh
Black-Scholes 1D – European call
Burgers 1D – Raissi et al.
Burgers 1D – RAR
Diffusion 1D
Diffusion 1D – hard IC and BCs
Diffusion 1D – resampling
Diffusion-reaction 1D
Fractional diffusion 1D
Heat 1D – Dirichlet
Heat 1D – resampling
Klein-Gordon 1D
Schrödinger 1D
Wave 1D – space-time multi-scale Fourier features
Diffusion 1D – unknown coefficient
Diffusion-reaction 1D – unknown k_f, D
γ_2, γ_1
λ_1, λ_2
D
a
γ
c
C
C
D, k_f
2D, steady
Fractional Poisson 2D – disk
Helmholtz 2D – Dirichlet
Helmholtz 2D – hyperparameter search
Helmholtz 2D – sound-hard scattering
Helmholtz 2D – square with hole
Laplace 2D – disk
Linear elasticity 2D – plate
Linear elasticity 2D – plate, soft BCs
Navier-Stokes 2D – Kovasznay flow
Poisson 2D – disk
Poisson 2D – L-shape
Fractional Poisson 2D – unknown order α
k_0
μ, λ
Re
f
α
2D, time-dependent
Allen-Cahn 2D – Wight & Zhao
Allen-Cahn 2D – Mattey & Ghosh
Cahn-Hilliard 2D – Wight & Zhao
Cahn-Hilliard 2D – Mattey & Ghosh
Burgers 2D – Mathias et al.
Heat 2D
λ, $\lambda\varepsilon^2$
c_2, c_1^2
ε^2
κ, α
ν
α
3D
Fractional Poisson 3D – sphere
Poisson 3D – cube with spherical cut-out
Poisson 3D – sphere
Allen-Cahn 3D – Wight & Zhao
Heat 3D
Navier-Stokes 3D – Beltrami flow
f
$\lambda\varepsilon^2$, λ
α
ν
10^{-6} 10^{-5} 10^{-4} 10^{-3} 10^{-2} 10^{-1} 10^{0}
Relative L_2 error
at the final reference time
10^{-3} 10^{-1} 10^{1}
Parameter or field
error (%)